



Validating alternative methodologies to estimate the regime of temporary rivers when flow data are unavailable

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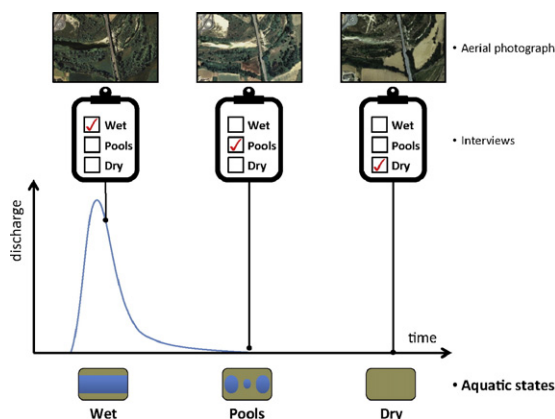
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HIGHLIGHTS

- The regimes of temporary rivers control the temporal patterns of aquatic life.
- As most temporary rivers are ungauged other methods than flow records are needed.
- Interviews and aerial photographs series were tested versus flow records.
- Both methods are best used together, giving information additional to flow records.
- Temporary aquatic states must be noted down when biological samples are taken.

GRAPHICAL ABSTRACT



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ABSTRACT

Hydrological data for assessing the regime of temporary rivers are often non-existent or scarce. The scarcity of flow data makes impossible to characterize the hydrological regime of temporary streams and, in consequence, to select the correct periods and methods to determine their ecological status. This is why the TREHS software is being developed, in the framework of the LIFE Trivers project. It will help managers to implement adequately the European Water Framework Directive in this kind of water body. TREHS, using the methodology described in Gallart et al. (2012), defines six transient 'aquatic states', based on hydrological conditions representing different mesohabitats, for a given reach at a particular moment. Because of its qualitative nature, this approach allows using alternative methodologies to assess the regime of temporary rivers when there are no observed flow data. These methods, based on interviews and high-resolution aerial photographs, were tested for estimating the aquatic regime of temporary rivers. All the gauging stations (13) belonging to the Catalan Internal Catchments (NE Spain) with recurrent zero-flow periods were selected to validate this methodology. On the one hand, non-structured interviews were conducted with inhabitants of villages near the gauging stations. On the other hand, the historical series of available orthophotographs were examined. Flow records measured at the gauging stations were used to validate the alternative methods. Flow permanence in the reaches was estimated reasonably by the interviews and adequately by aerial photographs, when compared with the values estimated using daily flows. The degree of seasonality was assessed only roughly by the interviews. The recurrence of disconnected pools was not detected by flow records but was estimated with some divergences by the two methods.

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The combination of the two alternative methods allows substituting or complementing flow records, to be updated in the future through monitoring by professionals and citizens.

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1. Introduction

Temporary rivers are water courses that undergo the recurrent interruption of flow or even the complete drying of their channel. Temporary streams are not only prevalent in dry climate areas (e.g. [Rossouw et al., 2005](#); [Levick et al., 2008](#)), but also make up the first-order river network in many drainage basins in wetter climates ([Fritz et al., 2006](#)). [Nadeau and Rains \(2007\)](#) estimated that this kind of river accounts for 59% of total river length in the United States, excluding Alaska. The occurrence of these rivers is expected to increase in the near future because of both climate warming and rising water consumption due to human activities ([Tooth, 2000](#); [Larned et al., 2010](#)). In the last decade there has been increasing concern about temporary rivers in the scientific community, which is compensating for a long period of disregard ([Datry et al., 2011](#); [Leigh et al., 2015](#)). The prime causes for this interest are the need for more ecologically sound management of waters (e.g. [Nikolaidis et al., 2013](#)), the requirements for implementing environmental regulations (e.g. [Nadeau and Rains, 2007](#)) and scientific curiosity for a poorly known domain.

An outstanding characteristic of temporary rivers is the particularity of the aquatic life they accommodate. The structure and composition of biological communities in temporary river reaches are adapted to the temporariness of the habitats and are transient, as they are strongly dependent on the temporal changes in the aquatic habitats determined by the hydrological conditions (e.g. [Larned et al., 2010](#); [Schriever et al., 2015](#)). Therefore, aquatic fauna biological indicators cannot be used to assess the ecological quality of a temporary river reach without taking into account the controls imposed by the hydrological regime ([Prat et al., 2014](#)). For this reason, the biological communities in reference temporary rivers, as well as the sampling calendar, must be suited to the river regime ([Cid et al., 2016](#)).

[Gallart et al. \(2012\)](#), incorporating former attempts ([Boulton, 2003](#); [Fritz et al., 2006](#)), schematised the relationships between the different hydrological phases and the corresponding habitats for aquatic invertebrates in temporary rivers. These authors put forward a description of the river regime based on a sequence of 'aquatic states'. These states summarize the set of mesohabitats occurring on a given reach at a particular moment, depending on the hydrological conditions. Six aquatic states were defined from wet to dry: flooding conditions (*Hyperheic*), full prevalence of all the possible mesohabitats (*Eurheic*), sequence of pools connected by flowing water threads (*Oligorheic*), occurrence of isolated pools (*Arheic*), disappearance of surface water, with the wet alluvium still allowing underground aquatic life (*Hyporheic*) and the disappearance of any active aquatic habitat in the river bed (*Edaphic*). In the long term, the river regime is characterized by the temporal pattern of the relative frequencies of the diverse aquatic states, allowing the probability of occurrence of every aquatic state in every month to be determined. Two complementary regime metrics based on the statistics of zero flows ([Gallart et al., 2012](#)) were used to assist the classification and comparison of temporary river regimes: flow permanence (Mf) and seasonal predictability of periods without flow (Sd₆).

The LIFE TRIVERS (LIFE13 ENV/ES000341) project was launched in 2014 with the main objective of implementing the developments made by [Gallart et al. \(2012\)](#), as well as the concomitant tools developed by [Prat et al. \(2014\)](#), for managing temporary rivers within the regulations of [Water Framework Directive \(2000\)](#). A major target of this project is the development and test of a software tool (TREHS, Temporary Rivers' Ecological and Hydrological Status) designed to assist the correct operational implementation of the Water Framework Directive

(WFD) in this kind of water body. As temporary rivers are typically ungauged, or poorly gauged, the main problem found is the lack of adequate historical data for determining river regimes. Therefore, alternative methods, such as interviews or the interpretation of aerial photographs, were used.

In hydrology and other earth sciences, the use of interviews to collect qualitative or quantitative information, something normal in social or medical sciences, is an uncommon practice. The advantage of this method is that it allows community knowledge to be added to scientific research and management know-how. Local people may have much greater knowledge than outsiders of the local environment, in particular some characteristics that help to define river regimes. However, while the use of interviews can be useful in areas where information is scarce and the phenomena studied are observable ([Calheiros et al., 2000](#)), little information exists for the inclusion of local knowledge in hydrological studies, although this method is becoming widespread particularly in resources management and restoration (e.g. [Leauthaud et al., 2013](#); [O'Keefe et al., 2015](#); [Westling et al., 2014](#)), as well as in flood risk assessment (e.g. [Miceli et al., 2008](#); [Box et al., 2016](#)).

Remote sensing, because of its low cost and high sequence rate, has been used for monitoring water bodies (e.g. [Arledler et al., 2010](#)). In particular, [Callow and Boggs \(2013\)](#) used the Moderate Resolution Imaging Spectroradiometer (MODIS) sensor with high temporal frequency (twice daily) and 250 m of spatial resolution to derive five flow types and six hydrological metrics in an entire 1996 km² dryland area in northern Australia. Unfortunately, this resolution is not at all suited to the width of temporary rivers in Mediterranean Europe, where small streams have summer active channels that are just a few metres wide. Alternatively, high-spatial-resolution aerial photographs freely available from global providers (Google Earth) or local ones (e.g. IGN or ICGC in Spain) may be used to estimate aquatic states at the test river reaches on the capture dates, although dense vegetation cover may impede channel inspection ([Turner and Richter, 2011](#)).

Other methods such as those tested by [Nadeau et al. \(2015\)](#), based on field geomorphic, hydrologic and biologic attributes, as well as hydrographic datasets coupled with synthetic stream networks, might also be applicable if they could be modified for assessing the occurrence of the aquatic states.

Besides these methods, designed for assessing past middle-term stream regimes, a number of alternative methods have been recently developed for monitoring the occurrence of water in given points of the stream network using low-cost temperature and/or electrical conductivity sensors (e.g. [Ronan et al., 1998](#); [Goulsbra et al., 2009](#); [Chapin et al., 2014](#)). Furthermore, there is a growing interest in the involvement of the general public in data collection, partly helped by new technologies ([Buytaert et al., 2014](#)). As an example relevant for temporary streams, the contribution of trained volunteers has been used to repeatedly map the expansion and contraction of the stream networks ([Turner and Richter, 2011](#); [Datry et al., 2016](#)). In France, this method is used for regularly mapping the continuity of stream flow in several catchments (<http://www.eau-poitou-charentes.org/Suivi-assecs-Federations-de-Peche.html>), as well as for operating a national observatory of summer low flows (<http://www.reseau.eaufrance.fr/projet/observatoire-national-etiages>).

This is the background to the purposes of the present study: first, to present a novel alternative method, based on interviews and the interpretation of high-resolution aerial photographs, for estimating the middle-term temporal patterns of the aquatic states and the above-described metrics in ungauged temporary rivers. The second and third

purposes are to validate this method in river reaches with available flow gauging records and to highlight the strengths and limitations of the methods evaluated, with some perspective on their use.

2. Materials and methods

2.1. Study site

A total of 13 gauging stations belonging to the Catalan Internal Catchments (CIC), all with recurrent zero-flow periods, were selected for this study. These stations include from small- (70 km²) to medium-sized (900 km²) catchments where annual precipitation ranges from 500 to 900 mm, with semiarid to subhumid climates. Temporari-ness in these catchments is partly of natural origin, enhanced by under-ground and surface water abstractions for agricultural, urban and industrial uses. The gauging stations and the reaches near them were visited and the methods explained below were applied.

2.2. Methodology used with flow data records

Flow data from the gauging stations were provided by the Catalan Water Agency (ACA). The methodology described by Gallart et al. (2012) for analysing the temporal patterns of the aquatic states and finding the regime metrics, was adapted to the characteristics of the data when necessary. Average monthly values of flow were used, because this time scale was considered adequate for biological processes. However, daily flows were also analysed in the drier stations in order to test the validity of the monthly values. When possible, records from the 21st century were preferred to older records, in order to avoid possible temporal inhomogeneity. The flow series used were >10 years long, except in the station 13 where the record was 8 years long. Data gaps were not >10% of the record in any of the selected stations. The data were managed by the TREHS software (freely available at www.lifetrivers.eu). The TREHS software allowed us to (a) Calculate regime metrics and their statistics from flows and interviews; (b) Assess temporal patterns of the aquatic states and draw the corresponding graphs.

2.2.1. Calculating regime metrics from flows

The metrics used to characterize the aquatic regime of temporary rivers were based on the classification of flow values in the two categories, positive and zero flows, because the interruption of flow is considered the most relevant feature constraining aquatic life. The selected metrics were flow permanence (Mf) and the predictability of dry periods (Sd₆). Nevertheless, the possible incidence of short-lived events cast doubt on the adequacy of mean monthly flows for examining flow interruption. Events lasting a few days during a dry month would give positive monthly values even though the river remained dry most of the time. For this reason, both metrics were also calculated using daily flows for the sites with flow permanence smaller than 0.90.

The flow permanence Mf metric indicates the long-term mean annual relative number of months (or days) with flow (taking values between 0 and 1), whereas Sd₆ indicates the predictability of periods without flow and is calculated by Eq. (1):

$$Sd_6 = 1 - \left(\frac{\sum_{i=1}^6 Fd_i}{\sum_{j=1}^6 Fd_j} \right) \quad (1)$$

where Fd_i represents the multi-annual frequencies of zero-flow months for the contiguous 6 wetter months of the year and Fd_j represents the multi-annual frequencies of zero-flow months for the remaining 6 drier months. This metric is dimensionless and takes the value of 0 when zero flows occur equally throughout the year in the long run; and 1 when all the zero flows occur in the same 6-month period every year. In the case of permanent rivers without flow interruption, Sd₆ is set to 1.

Both the Mf and Sd₆ metrics were calculated for every year on record for obtaining their standard deviation. Standard deviations were preferred to standard errors of the means as a way of illustrating the temporal variability of the metrics.

2.2.2. Assessing temporal patterns of aquatic states from flows

The temporal patterns of the aquatic states were described by means of aquatic states frequency graphs (ASFG). These graphs (see Fig. 2 in Section 3.1) show the long-term relative frequency of the diverse aquatic states for every month in the year (Gallart et al., 2012).

The flow duration curves (distribution functions of the flow values) were used to determine the flow threshold values separating the aquatic states. This allowed the monthly frequencies of the obtained flow ranges to be converted into the temporal patterns of the aquatic states then shown in the ASFGs. Even if this is a rather subjective method, it is not a critical step because it is only used to build the diagram and not to calculate the metrics (Gallart et al., 2012).

The flow values corresponding to the Hyperheic state (flooding) were taken as those occurring <5% of the time. The flow thresholds between the Eurheic (abundant riffles) and Oligorheic (pools connected by thin water threads) states were set at the recurrent downward bends of the curve for low flows. Finally, as all the gauging stations studied measure flow over alluvium mantles, allowing the occurrence of disconnected pools during zero-flow records, the thresholds between Oligorheic and Arheic (disconnected pools) and drier states were set at zero flows.

The resulting ASFGs had, therefore, just four classes (colours), the first three corresponding, from the wetter to the drier, to the Hyperheic, Eurheic and Oligorheic states, and a last one grouping the three remaining ones: Arheic, Hyporheic and Edaphic.

2.3. Methodology used with interviews

Interviews were conducted with neighbours of the fluvial system, i.e. people living near the rivers or working in vegetable gardens on the river sides. Interviewees were selected haphazardly from people who were near the river and were willing to answer the interview. The interviews lasted until speech saturation was reached (Glaser and Strauss, 1967) and included open-ended questions used to collect the information necessary to assess the seasonal patterns of the aquatic states and to calculate the Mf and Sd₆ metrics, as well as other complementary information about the current and historical river regime. The interview form (see Appendix A) was not given to the interviewee, but was filled in by the interviewers during or immediately after the interview. The interviews were carried out between November 2014 and November 2015, always by the same interviewers.

The number of interviewees per river followed the concept of sample size saturation, which is that the collection of new interviews does not provide any further information on the issue studied (Mason, 2010). However, for several reasons our sample size saturation level (for each river) was much lower than recommended for qualitative studies (e.g. Guest et al., 2006). Indeed, since this was a pilot study, designed for evaluating the proposed methodology for its operational implementation in a large number of river reaches, we prioritized a sample that was representative of the hydrologic variability observed in the area, rather than a sample with many interviews performed for a few rivers.

From a total of 56 interviewees 83% were inhabitants of the villages crossed by the rivers studied, the remaining (17%) were persons that worked in small vegetable gardens located at the riverside. The majority of the interviewees were male (87%), with a mean age of 65 years and who lived >43 years near the river.

To calculate the metrics from interviews and derive the ASFGs, given the impracticality of obtaining information from this methodology for each month and for all the aquatic states described, the months were grouped by seasons and the aquatic states were reduced from six to

three. Thus, the core of the questionnaire consisted of a template with four columns corresponding to the four seasons and three rows corresponding to three condensed aquatic states (flow, disconnected pools and dry river bed). Each box represents the number of months of occurrence of the corresponding state in the corresponding season, so every column must cover 3 months and every row up to 12 months (see Appendix A).

Furthermore, the interviewees were asked about the occurrence of wet/saturated alluvium after the disappearance of surface water (pools). Yet, for the driest rivers, when the interviewee indicated that the river flow 'ceases for long periods', the question was reformulated in a converse way, asking 'How many days per year does the river carry water?'.

2.3.1. Calculating regime metrics from interviews

With the information described above, for every interview, M_f was calculated as the ratio from the sum of the months with 'flow' row to 12. In the case of values smaller than one month per year, the ratio from the days with flow to 365 was used instead. The statistics of the M_f values for all the interviews in the reach were then calculated and those interviews whose M_f was an outlier were discarded.

The metric S_{d6} was calculated by differentiating the semester with fewer dry months (S_{wet}) from the semester with more dry months (S_{dry}), and then applying Eq. (2):

$$S_{d6} = 1 - \left(\frac{S_{wet}}{S_{dry}} \right) \quad (2)$$

After calculating these metrics, the set of interview results for every site and metric was described by the mean and the standard error of the mean, as here interest lay in the estimated value of the metric instead of the variability of the interviews.

2.3.2. Using interviews to assess the temporal patterns of aquatic states

The ASFGs derived from interviews had to be simplified according to the available information. Three classes (colours) were used, corresponding to flow (Hyperrheic, Eurheic and Oligorheic states), disconnected pools (Arheic state) and dry river bed (Hyporheic and Edaphic states) conditions. The temporal scale was reduced from twelve months to four seasons, as well.

This means that the comparison between ASFGs derived from measured flows and interviews was not straightforward because the categories did not coincide. The only coincidence between the two groups of graphs was the boundary corresponding to zero flows, i.e. between the Oligorheic state and the class grouping the drier states, in the graphs derived from flows and between 'flow' and 'isolated pools' conditions in the case of interviews.

2.4. Methodology used with aerial photographs

Series of about 12 orthophotographs with a resolution between 0.25 and 0.50 m and with a repetition rate of almost one year were obtained for every reach from the web site of the Cartographic and Geological Institute of Catalonia (ICGC). The images were inspected to identify three different hydrological conditions in the reaches near the gauging stations: continuous flow, disconnected pools and dry river bed (Fig. 1). Unfortunately, the exact capture dates of each image could not be obtained from the provider, because these are mosaics of photographs taken on diverse dates in the same year.

The resulting information allowed estimation of flow permanence (M_f) as the relative frequency of continuous flow, as well as the frequency of disconnected pools (Arheic state). The measurement of dispersion used in this case was the resolution of the estimate, calculated as the inverse of the number of inspected images. The seasonal patterns necessary for calculating the S_{d6} metrics and drawing the ASFG could not be obtained due to the limited number of observations and the lack of capture dates.

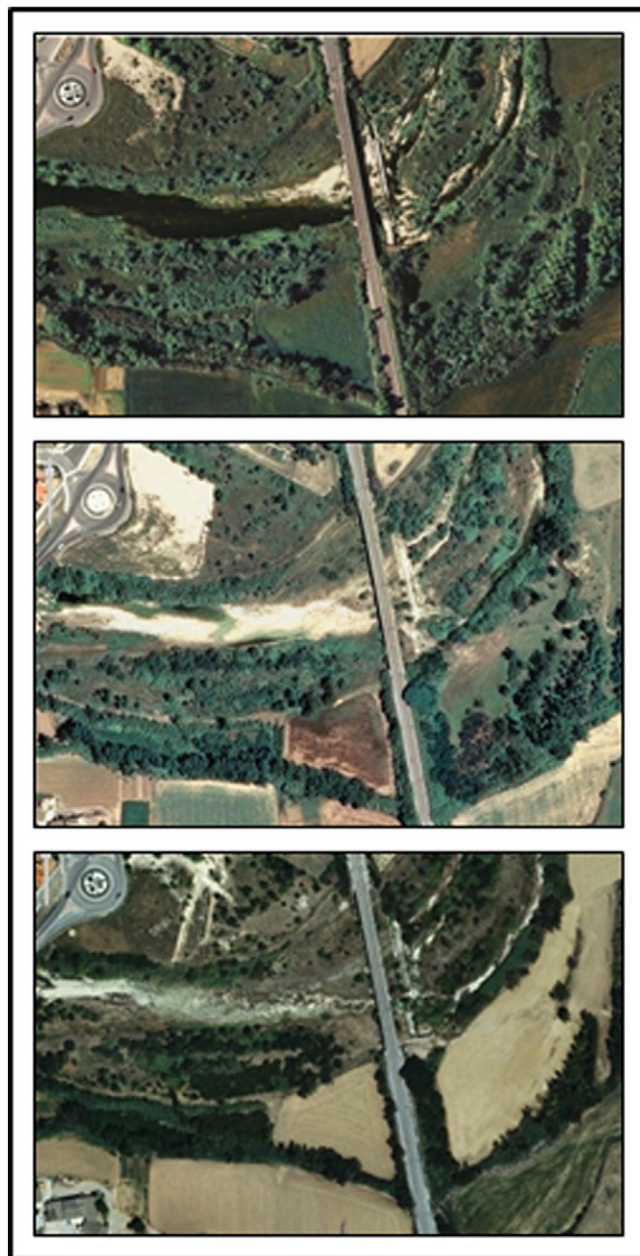


Fig. 1. Sequence of aerial orthophotographs of the stream Manol at the Santa Llogaia gauging station. The upper image shows continuous flow, the middle one disconnected pools and the lower one dry stream bed. The width of the images is 500 m. Source: ICGC.

3. Results and discussion

Table 1 shows the river reaches near gauging stations included in the study along with the main results obtained. They embrace from nearly permanent to ephemeral rivers that bear water <25% of the time. Daily flows and aerial photographs were used for assessing the regime metrics only for sites with flow permanence (M_f) <0.90. The results of this table are analysed in Section 3.2 (Figs. 4 to 7).

3.1. Temporal patterns of aquatic states

Fig. 2 shows the ASFGs generated for the study sites with the mean monthly flows, ordered from the more permanent to the less permanent. Many of them show marked seasonality, with the drier periods located in summer, although some of them show a rather flat pattern

Table 1

Gauging stations examined in this study and metrics obtained. The numbers in the first column are used to label the stations in the figures. Longitude and latitude are in degrees (ETRS89). Mf and Sd₆ are dimensionless ratios. Appendix B encloses a map with the locations.

	Gauging station name	Longitude	Latitude	Mf				Sd ₆	
				Flows		Interviews	Photos	Flows	Interviews
				Monthly	Daily				
1	Ripoll at Castellar del Vallès	2.08112	41.60865	0.99		0.99		1.00	1.00
2	Mogent at Montornès	2.26520	41.54665	0.98		1.00		0.89	1.00
3	Onyar at Girona	2.82497	41.97490	0.96		0.98		1.00	1.00
4	Tenes at Lliçà de Vall	2.24566	41.58650	0.95		0.88		0.85	1.00
5	Llobregat de Muga at Peralada	3.00878	42.30420	0.94		0.84		0.68	1.00
6	Gaià at Vilabella	1.35677	41.26720	0.94		0.86		1.00	1.00
7	Françolí at Tarragona	1.23629	41.14868	0.90	0.77	0.83	0.85	0.97	1.00
8	Tordera at Fogars	2.70796	41.71490	0.88	0.78	0.71	0.86	1.00	1.00
9	Riera de Caldes at Sta. Perpètua	2.19773	41.52254	0.75	0.56	0.37	0.31	0.83	0.42
10	Daró at Serra de Daró	3.07127	42.02501	0.72	0.59	0.53	0.55	0.37	0.82
11	Manol at Sta. Llogaia	2.96892	42.23790	0.66	0.57	0.64	0.62	0.26	0.69
12	Ridaura at Sta. Cristina d'Aro	2.97943	41.81810	0.49	0.50	0.19	0.50	0.54	0.35
13	Daró at La Bisbal	3.03816	41.96460	0.23	0.20	0.13	0.18	0.24	0.18

(Daró, at both locations, and Manol). The Ridaura river shows a pattern characterized by a wetter period in spring, attributable to the rainfall regime in the littoral area where it is located.

Fig. 3 shows the ASFGs obtained by averaging the interviews for each site. To compare these ASFGs with those obtained with the results of flow records, it must be taken into account that the lower class in Fig. 2 (orange) corresponds to the two lower classes in Fig. 3 (yellow and red).

At first glance, the interviews provided simplifications of the real temporal patterns of flows. Nevertheless, some clear differences appear. Seasonality was exaggerated in some cases (Llobregat de Muga, Daró at Serra de Daró and Manol), whereas in other cases it was underestimated (Riera de Caldes and Ridaura). Furthermore, at Ridaura, the main wet period occurring in spring was wrongly attributed to winter by the interviewees. Unfortunately, as indicated in Section 2.4, the data obtained from aerial orthophotographs were not suitable for analysis of the temporal patterns of the aquatic states.

3.2. Temporary regime metrics

The comparison between the flow permanence metrics (Mf) calculated using averaged annual flows and daily flows is shown in Fig. 4. As presumed, the effect of short-lived events appears to be more relevant when averaged to the full month for some gauging stations (i.e. Riera de Caldes). Subsequently, for all the stations with Mf < 0.90, daily flows were used to obtain the regime metrics from flow records.

3.2.1. Comparison between results obtained with flow records and interviews

The comparison between the flow permanence metric (Mf) obtained from the flow records and from the interviews is shown in Fig. 5. This graph shows a scattered error around the line 1:1 for the wetter sites (Mf higher than 0.6) and a systematic bias towards underestimation of flow permanence by the interviews for the drier sites. The scatter was somewhat compensated for by the large inter-annual variability of the metric. Mf calculated from monthly averaged flows is also shown (grey dots in Fig. 5). The latter shows greater deviations from the 1:1 line.

The comparison of the seasonal predictability metrics (Sd₆) obtained with flow records and with interviews was less satisfactory. As is shown in Fig. 6, interviews clearly overestimated seasonality at Llobregat de Muga, Daró at Serra de Daró and Manol (sites 5, 10 and 11, respectively) and underestimated it at Riera de Caldes and Ridaura (sites 9 and 12). These differences are consistent with those comparing the ASFGs calculated by both methods (Figs. 2 and 3). Since maximum Sd₆ decreases

with decreasing flow permanence for Mf lower than 0.5 (Gallart et al., 2012), the underestimation of low values of Mf by the interviewees for sites 9 and 12 (Fig. 5) implies some reduction of the corresponding Sd₆ values estimated by the interviews.

It may be assumed that the quality of the results was conditioned by limited knowledge of the river regime of the interviewees, so these results could be improved if these were made to key informants who had a more expert professional or leisure relationship with the river (e.g. personnel of the water authorities, anglers).

3.2.2. Comparison between results obtained with flow records and aerial photographs

The comparison between the Mf metrics obtained with flow records and aerial photographs is shown in Fig. 7. All the sites show great consistency between the two estimates, given the considerable uncertainty of the estimates from flow records, with the exception of the Riera de Caldes (site 9), where aerial photographs clearly underestimated flow permanence. Of the sites studied, this gauging station shows the strongest prevalence of flow interruption during summer (Fig. 2). Therefore, a possible cause of the bias of the results obtained with aerial photographs could be the month of the year when most of the photographs were taken in this area. However, as there is no information on the dates when photographs were taken, it is not possible to validate this hypothesis.

3.2.3. Comparison between results obtained with interviews and aerial photographs

Measured flows, the data used in this study to validate the two other methods, do not provide information on the occurrence of disconnected pools when surface flow is practically zero. In temporary rivers, the Arheic state (isolated pools) is crucial for the maintenance of aquatic life during periods without surface flow, particularly for fish and other aquatic communities with no aerial dispersal (Bond and Cottingham, 2008). Moreover, biological quality assessments using available standardized methods for determining the ecological status can only be applied when stream mesohabitats are connected (Bonada et al., 2006; Buffagni et al., 2010; Prat et al., 2014). Therefore, obtaining information on the frequency of the Arheic state can be decisive but it is not available from flow data. Fortunately, both interviews and aerial photographs allowed us to investigate the rate of recurrence of the Arheic state.

Fig. 8 shows the comparison between the frequency of the Arheic state determined from interviews and aerial photographs. Both the reaches with the highest recurrence of pools (sites 10 and 11) and those with the lowest (sites 4, 7, 8 and 12) showed acceptable consistency between the two estimates, if the associated uncertainty is

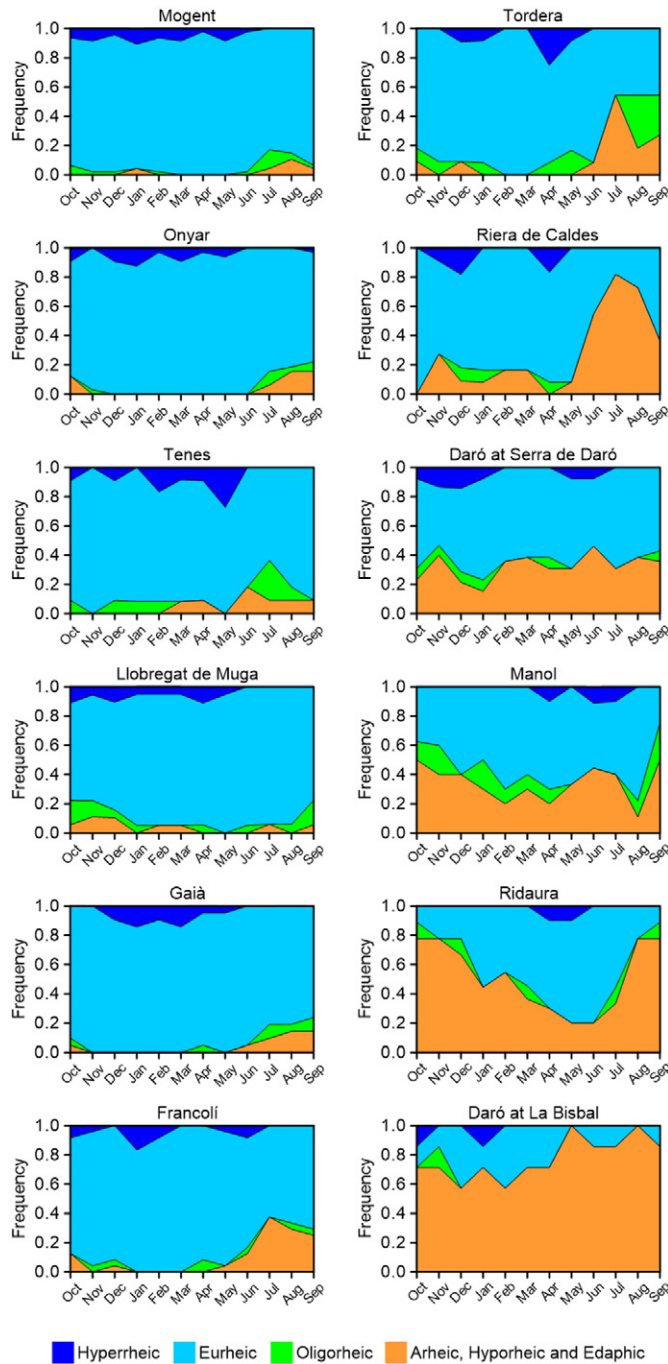


Fig. 2. Aquatic state frequency graphs obtained from the flow records at the gauging stations. The graph of the Ripoll river is not shown because it is very similar to the Mogent river graph.

taken into account. Sites 5 and 6 are reaches with high flow permanence ($M_f = 0.94$), where just one image per year could not detect possible Arheic periods. Conversely, sites 9 and 13 are reaches with lower flow permanence (M_f equal to 0.56 and 0.20, respectively), where the occurrence of periods with pools was noticed in the aerial photographs but not or hardly mentioned by the interviewees.

Therefore, the low repetition rate of the used aerial photograph series not only prevented the assessment of the temporal patterns of the aquatic states, but also limited the quality of their averaged values. The use of higher repetition rate remote imagery, such as the one

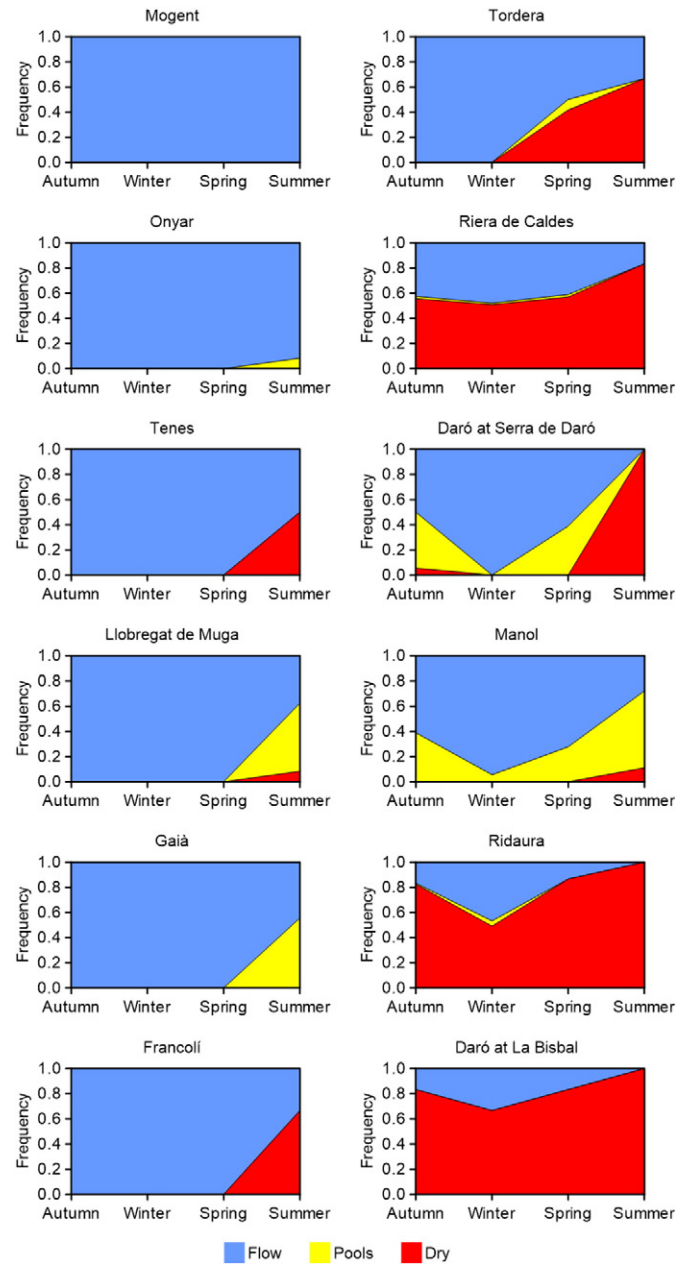


Fig. 3. Aquatic state frequency graphs obtained from the results of the interviews. The graph of the Ripoll river is not shown because it is very similar to the Mogent river graph.

provided by the QuickBird satellite (e.g. Whited et al., 2013), may solve this problem though notably increasing the costs.

4. Concluding remarks

To test methods that are suited to analysing the medium-term regime of temporary rivers in ungauged catchments, 13 temporary river reaches near gauging stations in the Catalan Internal Catchments were investigated. The method used as a reference was flow records measured at the gauging stations, whereas the alternative methods were interviews with local inhabitants and the inspection of series of annual high-resolution aerial photographs.

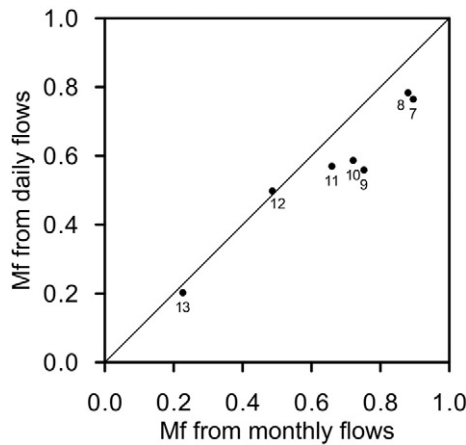


Fig. 4. Comparison between the flow permanence Mf values calculated from averaged monthly and daily flows. The line shows the 1:1 relationship. The correlation coefficient is significant ($r = 0.981$, $p < 0.001$). The numbers correspond to the sites shown in Table 1.

With flow records the monthly patterns of four aquatic states could be determined: Hyperheic (flooding), Eurheic (abundant flow), Oligorheic (scarce flow) and the lack of surface flow. The seasonal patterns of three conditions could be calculated from interviews: water flow, disconnected pools (Arheic state) and dry river bed. Examination of aerial photographs allowed the calculation of the frequency of these last three conditions, but the low recurrence rate, only one photograph per year, did not let us estimate the temporal patterns. Moreover, with this methodology, the inspection of some small headwater streams was difficult due to the dense vegetation cover. For that reason, open areas such as fords or weirs were there used for identifying the occurrence of flow.

Flow permanence in the reaches (flow relative duration) was reasonably estimated by the interviews and adequately by aerial photographs, when compared with the values estimated using daily flows. Nevertheless, interviews underestimated the flow permanence at gauging stations with values lower than 0.5, whereas aerial photographs seemed to underestimate flow permanence at a gauging station where most periods without flow occurred in summer. The comparison of

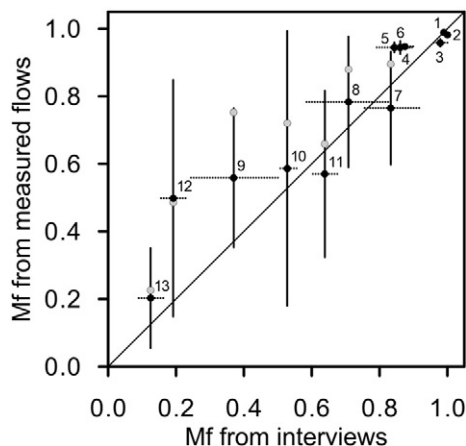


Fig. 5. Comparison between the flow permanence Mf values obtained with flow records and with interviews. The error bars of flow results (solid lines) represent the standard deviation and those of the interview results (dashed lines) represent the standard error of the mean. Grey dots represent the Mf values obtained with monthly averaged flows (error bars not shown). The line represents the 1:1 relationship. Both correlation coefficients are significant (black dots: $r = 0.946$, $p < 0.001$; grey dots: $r = 0.929$, $p < 0.001$). The numbers correspond to the sites shown in Table 1.

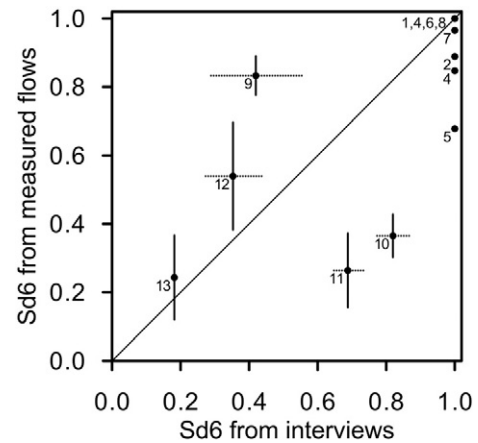


Fig. 6. Comparison of the 6-month predictability of dry period Sd₆ values obtained with flows and interviews. The error bars of flow results (solid lines) represent the standard deviation and those of the interviews results (dashed lines) represent the standard error of the mean. The line shows the 1:1 relationship. The correlation coefficient is marginally significant ($r = 0.626$, $p = 0.022$). The numbers correspond to the sites shown in Table 1.

the recurrences of the Arheic state (disconnected pools) between interviews and aerial photographs was mediocre. Both the degree of seasonality and the temporal patterns of flow interruption were just coarsely assessed by interviews. However, and taking into account the problems encountered with them, these alternative methodologies provided information impossible to acquire with flow records.

Therefore, it can be concluded that interviews, as reported in this study, were adequate to provide a first-order assessment of the regime of the aquatic states in the investigated reaches, although the information on the temporal patterns was not really reliable. In our opinion, the results obtained with interviews could be improved if these are with key informants who have a more expert professional or leisure relationship with the river regime. The examination of series of aerial photographs provided satisfactory information, although the low repetition rate did not allow assessment of the temporal patterns. The use of higher repetition rate remote imagery, at least one quarterly image may solve this problem. Anyway, the combination of these two alternative methods afforded more robust information.

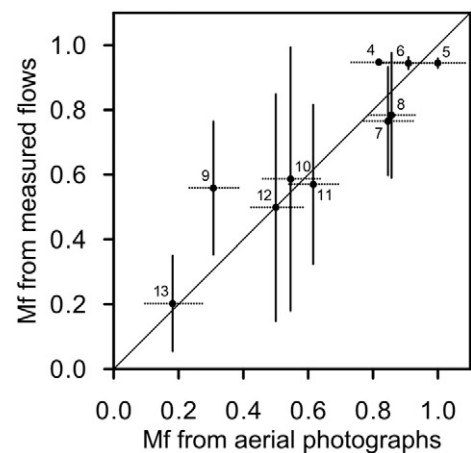


Fig. 7. Comparison between the flow permanence Mf values obtained with flows and aerial photographs. The error bars of flow results (solid lines) represent the standard deviation and those of the aerial photograph results (dashed lines) represent the resolution of the estimate. The line shows the 1:1 relationship. The correlation coefficient is significant ($r = 0.927$, $p < 0.001$). The numbers correspond to the sites shown in Table 1.

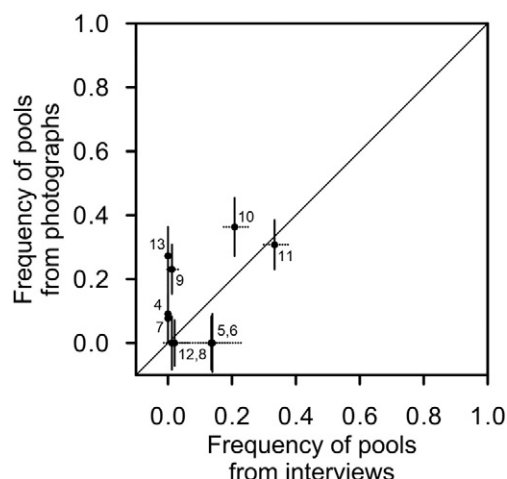


Fig. 8. Comparison between the relative frequencies of the Arheic state (disconnected pools) obtained with interviews and aerial photographs. The error bars of interview results (dashed lines) represent the standard error of the mean and those of the aerial photograph results (solid lines) represent the resolution of the estimate. The line shows the 1:1 relationship. The correlation coefficient is non-significant ($r = 0.413$, $p = 0.235$). The numbers correspond to the sites shown in Table 1.

Nevertheless, not one of the three methods on its own analyses adequately the statistics of the occurrence of the complete range of the six possible aquatic states, so historical information of their occurrence could not be obtained. Thus, this information can only be obtained in the future by systematic reporting of aquatic states, for example when water and biological samples are regularly taken by professionals in

surveillance networks or by observations reported by trained citizens; instrumented points and aerial photographs being necessary for temporal and spatial integration of the data obtained by the observers.

Involving citizens in hydrological and environmental surveillance has many advantages in terms on temporal and spatial coverage of observations, as well as on public engagement in environmental problems and solutions (Turner and Richter, 2011; Buytaert et al., 2014). New technologies make the scientific contribution of non-professional citizens easier and more reliable. For example,

smartphones are already in use by citizens for monitoring stream discharges (smartphones4water, available at <http://www.smartphones4water.org/s4w/smartphones4water/>) as well as for monitoring the quality of stream waters following the determination of key macroinvertebrates (riu.net, available at <http://www.ub.edu/fem/index.php/es/riunetini-ci-es>). Within the LIFE TRIVERS project, a smartphone application, containing some of the features of the TREHS software, is being developed for public use.

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Appendix A. Form used for the interviews

**a) General Information**

Date:	Interviewer:
Interviewee	
Name:	Domicile:
Age:	Relationship with the river:
Profession:	Years in contact with the river:
Catchment:	Water body:
Localization (GPS), X:	Interview number:
Y:	

b) Information about the general characteristics of the regime:

(1) In this river:

Flow never ceases	☐	[Move to part C]
Occasionally ceases (1 to 3 months)	☐	
Ceases often (between 3 and 9 months)	☐	
Ceases for long periods (>9 months)	☐	How many days does carry water?

(2) During the months when water does not flow, Are there still pools? How many days they last?

yes	no	days

(3) When there is no surface water, is there water in the alluvium?

yes	no

c) Information to estimate the metrics and the relative frequencies of the aquatic states

Fill the table considering for each season how many months the water flows, there are pools or the river is dry, using the following classification (use one decimal figure):

Very frequently (always or almost always)	3
Frequently (2 of the 3 months)	2
Occasionally (1 of the 3 months)	1
Rarely (about 15 days in the season)	0.5
Never (less than one week in the season)	0

The sum of each column must be 3 months

	Autumn	Winter	Spring	Summer
Flow				
Pools				
Dry				

**d) Changes in the regime**

If changes have been observed, indicate the related evidence and possible causes.

Evidence, for example, the appearance or disappearance of some animals (fishes, otters, etc) or changes in the water uses (i.e. pools where people swam, old laundry sinks in disuse, old (new) irrigation systems, etc).

Year	Observed change	Evidence	Possible cause
	discharge increase discharge decrease		
	discharge increase discharge decrease		
	discharge increase discharge decrease		

Appendix B. Supplementary data

Supplementary data associated with this article can be found in the online version, at <http://dx.doi.org/10.1016/j.scitotenv.2016.05.116>. These data include Google map of the most important areas described in this article.

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